

Evidence Gap Analysis: Assessing the Feasibility of a Fossil-Fuel-Free World Amidst Irrelevant Computational Literature

Input Question

Could we live in a fossil-fuel-free world?

Domain

Energy Science

Validation Status

needs_data

Problem Statement

The feasibility of transitioning to a fossil-fuel-free global society is constrained by complex technical challenges (e.g., renewable energy unpredictability, efficiency limits) and socio-economic hurdles. Current evidence suggests that while renewable sources like solar offer hope, they require significant technological advancements and policy support to replace the ubiquitous role of fossil fuels in energy and material production.

Rationale

The provided booklet excerpt highlights the tension between the necessity of phasing out fossil fuels and the current limitations of renewable technologies. However, the allowed evidence set consists entirely of computer science literature unrelated to energy systems. Therefore, this research plan focuses on validating the insufficiency of the current evidence base to answer the scientific question, establishing a rigorous knowledge gap rather than fabricating domain-specific conclusions.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Current evidence is insufficient to evaluate the feasibility of a fossil-fuel-free world; any specific technical or socio-economic claim regarding this transition currently lacks verifiable support within the allowed evidence set.
- **Mechanism**: The allowed evidence IDs (EV-Q115-d7477e1140f500c7a487e956, EV-Q115-c3108552764a5a0bce0f1cef, EV-Q115-24f15c9ea38132f3cfb95ea0, EV-Q115-04354aa7eab44d38a6da464a) pertain exclusively to software engineering, live coding, and programming environments. There is no semantic overlap with energy science, renewable technology, or fossil fuel systems. Therefore, no causal mechanism linking these sources to the research question can be constructed without violating factual grounding rules.

- ****Falsifiable Prediction****: If a comprehensive semantic search of the full text of all four allowed evidence cards yields zero mentions of 'energy', 'fossil fuel', 'renewable', 'grid', 'storage', or 'petroleum', then the hypothesis of evidence insufficiency is confirmed, and no domain-specific sub-hypothesis can be validly generated from this set.

- ****Required Observations****: Full-text semantic analysis of EV-Q115-d7477e1140f500c7a487e956 confirming absence of energy-domain terminology ; Full-text semantic analysis of EV-Q115-c3108552764a5a0bce0f1cef confirming absence of energy-domain terminology ; Full-text semantic analysis of EV-Q115-24f15c9ea38132f3cfb95ea0 confirming absence of energy-domain terminology ; Full-text semantic analysis of EV-Q115-04354aa7eab44d38a6da464a confirming absence of energy-domain terminology

- ****Risk of Being Wrong****: Low risk. The quoted texts in the evidence extraction explicitly describe software verification, virtual agents for coding, SWE-bench, and Smalltalk databases. The probability of hidden energy-science content in these computer science papers is negligible. However, if future re-extraction reveals mislabeled metadata, this assessment would require revision.

Hypothesis 2

- ****Hypothesis****: Methodologies from live programming verification (liveness feedback loops) could theoretically be adapted to model real-time grid stability in high-renewable energy systems, but no current evidence supports their direct application to fossil-fuel-free transition feasibility.

- ****Mechanism****: Hypothetical cross-domain transfer: Liveness verification in software (EV-Q115-d7477e1140f500c7a487e956) provides tight feedback between system state and user input. If analogous real-time feedback mechanisms were applied to power grid control systems, they might mitigate renewable unpredictability. However, this is purely speculative as no evidence links these CS concepts to energy infrastructure.

- ****Falsifiable Prediction****: If no peer-reviewed literature exists that applies 'liveness verification' or 'live programming feedback loops' to electrical grid stability or renewable integration, then this cross-domain hypothesis remains unsubstantiated and should be rejected as a viable research direction based on current evidence.

- ****Required Observations****: Literature search for 'liveness verification' AND 'power grid' OR 'renewable integration' ; Literature search for 'live programming' AND 'energy system modeling' ; Verification that EV-Q115-d7477e1140f500c7a487e956 does not contain energy applications

- ****Risk of Being Wrong****: High risk. This hypothesis extrapolates from unrelated domains without bridging evidence. It is included only to satisfy the requirement for a potentially simulatable/methodological hypothesis, but it is explicitly flagged as lacking evidentiary support.

Technical Details

This experiment is designed to validate the 'Evidence Insufficiency' hypothesis. The core technical task is a deterministic semantic audit of four specific Evidence Cards (EV-Q115-d7477e1140f500c7a487e956, EV-Q115-c3108552764a5a0bce0f1cef, EV-Q115-24f15c9ea38132f3cfb95ea0, EV-Q115-04354aa7eab44d38a6da464a). The system will perform full-text extraction and keyword matching against a predefined domain-specific lexicon related to energy systems (e.g., 'fossil fuel', 'renewable', 'grid', 'petroleum', 'energy transition'). The null result (zero matches) will statistically confirm that the provided evidence set is orthogonal to the research question, thereby validating the knowledge_gap classification.

Datasets

Source

```
```json
[
 {
 "id": "EV-Q115-d7477e1140f500c7a487e956",
 "type": "text",
 "description": "Full text of arXiv:1609.04233 regarding liveness in software verification."
 },
 {
 "id": "EV-Q115-c3108552764a5a0bce0f1cef",
 "type": "text",
 "description": "Full text of arXiv:2106.14835 regarding virtual agents in live coding."
 },
 {
 "id": "EV-Q115-24f15c9ea38132f3cfb95ea0",
 "type": "text",
 "description": "Full text of arXiv:2505.23419 regarding SWE-bench-Live and code agents."
 },
 {
 "id": "EV-Q115-04354aa7eab44d38a6da464a",
 "type": "text",
 "description": "Full text of arXiv:2309.11406 regarding Smalltalk database environments."
 }
]
```
```

Target

```
```json
{
```

```
"name": "Semantic Relevance Audit Report",

"format": "JSON",

"fields": [

"evidence_id",

"keyword_match_count",

"matched_terms",

"domain_classification"

]

}

'''
```

## Paper Abstract

Background: The transition to a fossil-fuel-free world faces significant technical and socio-economic hurdles, including the unpredictability of renewable energy sources. Method: This study evaluates the availability of verifiable scientific evidence regarding this transition by auditing a specific set of allowed literature. We perform a semantic relevance audit on four evidence cards derived from computer science domains (software verification, live coding). Validation Plan: We test the hypothesis that these sources contain no substantive information on energy systems by searching for domain-specific keywords. Results: Pending execution of the semantic audit experiment. The current analysis indicates a complete lack of relevant evidence, classifying the question as a knowledge gap requiring external data collection.

## Methods

1. Text Extraction: Retrieve full textual content from the provided URLs for all four Evidence IDs.
2. Lexicon Definition: Define a strict set of energy-domain keywords: ['fossil fuel', 'coal', 'oil', 'natural gas', 'petroleum', 'renewable energy', 'solar', 'wind', 'hydroelectric', 'nuclear', 'grid stability', 'energy storage', 'decarbonization'].
3. Semantic Search: Perform case-insensitive exact match and stem-based search for each keyword within each document.
4. Verification Logic: If total matches across all documents == 0, the hypothesis 'Current evidence is insufficient' is confirmed.

## Experiments

### Baselines

```
'''json

[
```

"Random Control: A random sample of 100 words from general news corpus to establish baseline frequency of energy terms (expected > 0).",

"Null Hypothesis Baseline: Assume all documents are relevant; expect high keyword density (>5%)."

]

...

## Metrics

```json

[

"Keyword Hit Rate: Number of unique energy-domain terms found per document.",

"Semantic Overlap Score: Cosine similarity between document embeddings and a standard 'Energy Transition' reference vector (expected ~0).",

"Relevance Classification Accuracy: Binary classification (Relevant/Irrelevant) based on hit rate threshold."

]

...

Ablation

Test sensitivity by expanding the keyword list to include broader terms like 'power', 'fuel', and 'resource' to ensure strict domain specificity is not masking subtle references.

Validation Protocol

Double-blind manual review of any false positives detected by the automated scanner. Cross-check with metadata titles and abstracts to ensure no mislabeled papers.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q115-d7477e1140f500c7a487e956** · arxiv · arXiv:1609.04233
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1609.04233.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=6119b55ccf7412ff224df17dfa5e0f83606ce6eff28b84eae28f1c25cc0cbad4

- ****EV-Q115-c3108552764a5a0bce0f1cef**** · arxiv · arXiv:2106.14835
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2106.14835.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=9d79da53ce8cc5a18c40bffa4d677c9f2fdb7087f9e5e0ea31d78f0fef6256c6
- ****EV-Q115-24f15c9ea38132f3cfb95ea0**** · arxiv · arXiv:2505.23419
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2505.23419.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:3|section:page-3|paragraph:1;
content_sha256=4126454bc9b188e92ec869b7ee7308e6469b46ae68feda9cc1c4790a9a940e72
- ****EV-Q115-04354aa7eab44d38a6da464a**** · arxiv · arXiv:2309.11406
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2309.11406.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=fed2e32b49ac11fb1f96e06396fdeaf5b1fe6f9c21fa7f47d98379b64f9aa92f

Reviewer Comments

- The candidate hypothesis correctly identifies a complete domain mismatch between the research question (fossil-fuel-free world) and the allowed evidence set (software engineering/live coding).
- The system appropriately classifies the situation as 'insufficient_evidence' rather than fabricating connections or hallucinating energy-related facts from CS papers.
- Experiment design is logically sound for verifying the meta-hypothesis of evidence irrelevance via semantic audit.
- Results field correctly states 'pending' and does not claim to have executed the verification experiment.
- No causal claims are made linking software verification concepts to energy systems; Hypothesis 1 is properly flagged as speculative and unsupported.
- All referenced evidence IDs exist in the provided catalog and are used only to demonstrate their irrelevance.

Revision History

Reproducibility Checklist

- Verify access to all four arXiv PDF URLs.

- Confirm exact version of the keyword lexicon used.
- Ensure text extraction handles PDF formatting artifacts correctly.
- Log all zero-match results for audit trail.